

$$\hat{s}(t) = \hat{s}_0 + \frac{t}{T} D \text{NN}(t, s_0, u), \quad D = \text{diag}\left(\frac{S_f T}{S_x}\right)$$

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \frac{1}{n} \sum \left\| \frac{\dot{\hat{s}} - f(\hat{s}, u)}{S_f} \right\|^2$$